Functions: Composite, Inverse & Logarithmic

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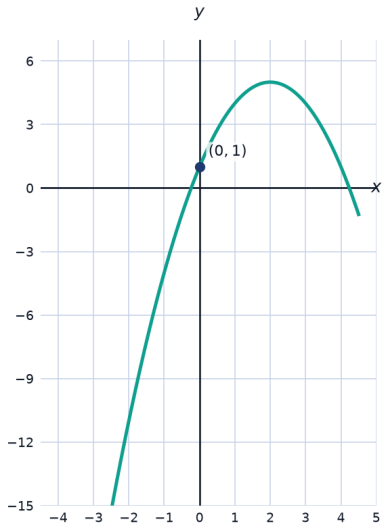
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Domain and range**

Find the domain of $f(x) = \frac{1}{x^2 - 9}$.

2 **[Foundational] • Target: Domain and range**

Find the range of $f(x) = -(x - 2)^2 + 5$.



3 **[Foundational] • Target: Composite functions**

Using the same f and g as the previous question, find $(g \circ f)(x)$ and state its domain.

4 **[Foundational] • Target: Verifying inverses**

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x - 1}{5}$ are inverses by showing $f(g(x)) = x$.

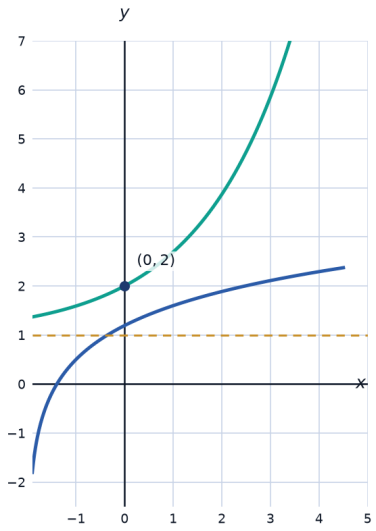
5 **[Foundational] • Target: Verifying inverses**

Does $f(x) = x^2$ (for all real x) have an inverse function? Explain.

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6 [Foundational] • Target: Logarithm laws

Expand $\log_b\left(\frac{x^3y}{z^2}\right)$ using the laws of logarithms.



7 [Foundational] • Target: Logarithm laws

Condense $2\log_5 x + 3\log_5 y - \log_5 z$ into a single logarithm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Logarithm laws

Evaluate $\log_2 32 - \log_2 4$ without a calculator.

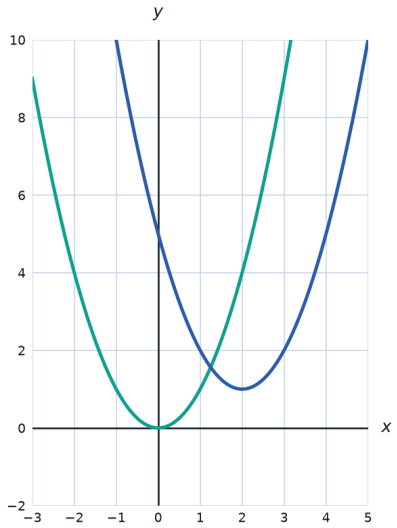
9 [Intermediate] • Target: Exponential equations and logs as inverses

The functions $f(x) = \log_3 x$ and $g(x) = 3^x$ are inverses. Use this fact to solve $3^x = 81$ without a calculator, and state the value of $g(f(16))$.

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10 [Intermediate] • Target: Transformations of functions

The graph shows $f(x) = x^2$ (teal) and $g(x) = (x-2)^2 + 1$ (indigo). Describe the transformation that maps f onto g.



11 [Intermediate] • Target: Transformations of functions

If $h(x) = -f(x+3)$ where $f(x) = x^2$, describe the transformations applied to f to obtain h.

12 [Intermediate] • Target: A well-designed word problem: investment growth and doubling time

This question has three parts. An investment fund grows according to $A(t) = A_0 e^{0.045t}$, where A_0 is the initial amount (in dollars) and t is time in years. (a) If $A_0 = 8000$ dollars, find $A(10)$ to the nearest dollar. (b) Find the time t, to the nearest tenth of a year, for the investment to double. (c) Solve for t in terms of A and A_0 — that is, find the inverse function $t(A)$.

13 [Intermediate] • Target: Verifying inverses

Given $f(x) = e^x$ and $g(x) = \ln(x+1)$, find the domain of $(g \circ f)(x)$ and evaluate $(g \circ f)(0)$.

14 [Intermediate] • Target: Logarithm laws

The function $f(x) = \frac{4x-7}{2x+8}$ has horizontal asymptote $y = H$ and vertical asymptote $x = V$. Determine the value of $H + V$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

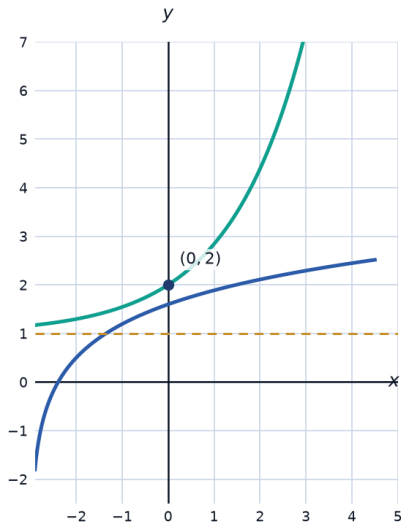
15 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.

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16 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 4.01 Lines, Triangles & Polygons

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



17 [Challenging] • Target: Transformations of functions • Cross-Lesson: 5.01 Trigonometry

An EST II Mathematics examination consists of 50 questions with an allocated testing time of 60 minutes. If a student completes the first 30 questions in 33 minutes, what is the maximum average time, in seconds, the student can spend on each of the remaining 20 questions?

18 [Challenging] • Target: A well-designed word problem: investment growth and doubling time • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A trigonometric function has the formula $f(x) = -3\sin\left(2x - \frac{\pi}{2}\right) + 4$. State its: (a) amplitude, (b) period, (c) phase shift, and (d) range.

19 [Challenging] • Target: Domain and range • Cross-Lesson: 7.01 Solid Geometry & Similarity

Five test scores are 70, 75, 80, 85, 90. Find the mean and range.

20 [Challenging] • Target: Composite functions • Cross-Lesson: 8.01 Statistics & Probability

[Test format & strategy · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Domain and range

What value must be excluded from the domain of $f(x) = \frac{1}{3x+3}$?

a. 3

b. -1

c. -4

d. 2

22 [Foundational] • Target: Composite functions

What is the smallest integer in the domain of $f(x) = \sqrt{2x+2}$?

a. -8

b. -9

c. -1

d. -10

23 [Foundational] • Target: Inverse functions

Consider the functions $f(x) = \frac{1}{x-2}$ and $g(x) = 3x + k$. If $f(g(x)) = \frac{1}{2x-1}$, find the value of k.



a. 3

b. 2

c. 1

d. -1

24 [Foundational] • Target: Verifying inverses

Let $f(x) = \log_2(ax + b)$ and $g(x) = 3^x - 1$. Given that $f(g(x)) = x$ and the point $(2, 1)$ lies on the graph of $y = g(x)$, find the value of $a + b$.

a. 4

b. 3

c. 2

d. 1

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25 [Foundational] • Target: Logarithm laws

Given the function $f(x) = 2x + 3$. What is the value of $f^{-1}(7)$?

a. 1

b. 2

c. 3

d. 4

26 [Foundational] • Target: Solving logarithmic equations

Consider the functions $f(x) = 3x + 2$ and $g(x) = x - 5$. What is the value of $f(g(3))$?

a. 1

b. -4

c. 13

d. -10

27 [Foundational] • Target: Exponential equations and logs as inverses

If $f(x) = -2x + 1$ and g is the inverse of f, find $g(f(-1))$.

a. 4

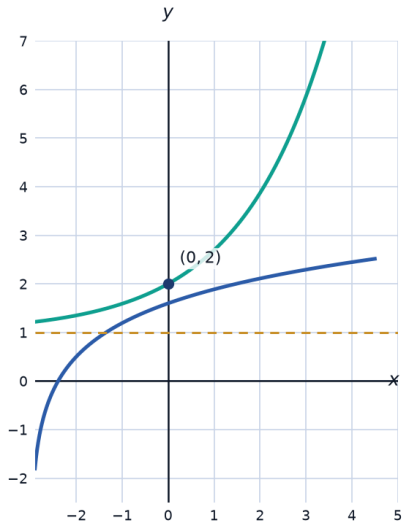
b. -1

c. 2

d. -10

28 [Foundational] • Target: Transformations of functions

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.



a. 8

b. 2

c. 4

d. 10

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: A well-designed word problem: investment growth and doubling time**

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. 0.0434

30 **[Intermediate]** • **Target: Domain and range**

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. 0.347

b. 0.416

c. 0.301

d. 0.477

31 **[Intermediate]** • **Target: Composite functions**

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

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32 [Intermediate] • Target: Inverse functions

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

c. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

33 [Intermediate] • Target: Verifying inverses

Which is the inverse of $f(x) = \log_2 x$?

a. $g(x) = \log_2 x$

b. $g(x) = 2x$

c. $g(x) = x^2$

d. $g(x) = 2^x$

34 [Intermediate] • Target: Logarithm laws

Use natural logs to solve $e^{x^2} = e^{3x+4}$.

a. $x = 4$ only

b. $x = -1$ only

c. $x = 3$

d. $x = 4$ or $x = -1$

35 [Intermediate] • Target: Solving logarithmic equations

Which function is the inverse of $f(x) = e^x$?

a. $g(x) = x^e$

b. $g(x) = \ln x$

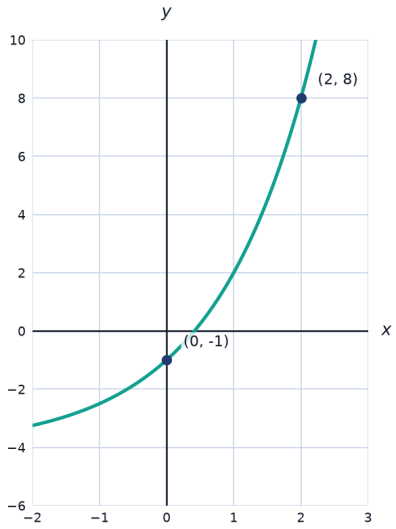
c. $g(x) = 10^x$

d. $g(x) = \log x$

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36 [Intermediate] • Target: Exponential equations and logs as inverses

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?



a. $-\frac{5}{2}$

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

37 [Intermediate] • Target: Transformations of functions

Find the range of the data set: $32, 35, 32, 11, 3, 3, 50$.

a. 21

b. 12

c. 18

d. 47

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: investment growth and doubling time • Cross-Lesson: 6.01 Circles & Coordinate Geometry

What is the sum of all real solutions to the equation $2^{\log_5(x^2)} - 6 \cdot x^{\log_5(2)} + 8 = 0$?

a. 30

b. 26

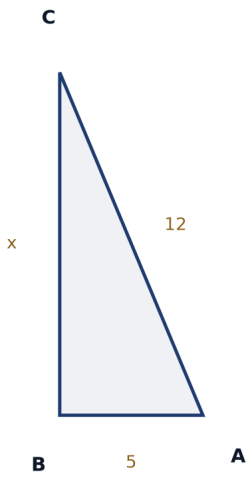
c. 6

d. 3

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39 [Challenging] • Target: Domain and range • Cross-Lesson: 7.01 Solid Geometry & Similarity

In a right-angled triangle ABC, angle B is 90°. If $AB = 5$ and $BC = 12$, what is the length of the hypotenuse AC?



a. 10

b. 11

c. 13

d. 17

40 [Challenging] • Target: Composite functions • Cross-Lesson: 8.01 Statistics & Probability

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. 20

b. 24

c. 32

d. 44

41 [Challenging] • Target: Inverse functions • Cross-Lesson: 1.01 Advanced Algebra

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19

b. 31

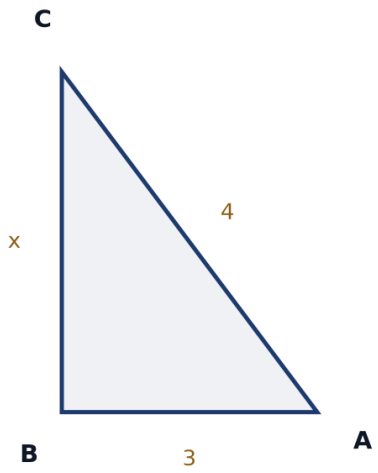
c. 16

d. 11

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42 [Challenging] • Target: Verifying inverses • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

In a right-angled triangle ABC, where angle B is 90 degrees, if AB = 3 and BC = 4, what is the length of the hypotenuse AC?



a. 7

b. 5

c. 12

d. 25

43 [Challenging] • Target: Logarithm laws • Cross-Lesson: 4.01 Lines, Triangles & Polygons

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$

b. $y = 0$

c. $y = 3$

d. $x = -1$

44 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 5.01 Trigonometry

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. a = 4, b = 3

b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

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45 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 6.01 Circles & Coordinate Geometry

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. 4

d. 6

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